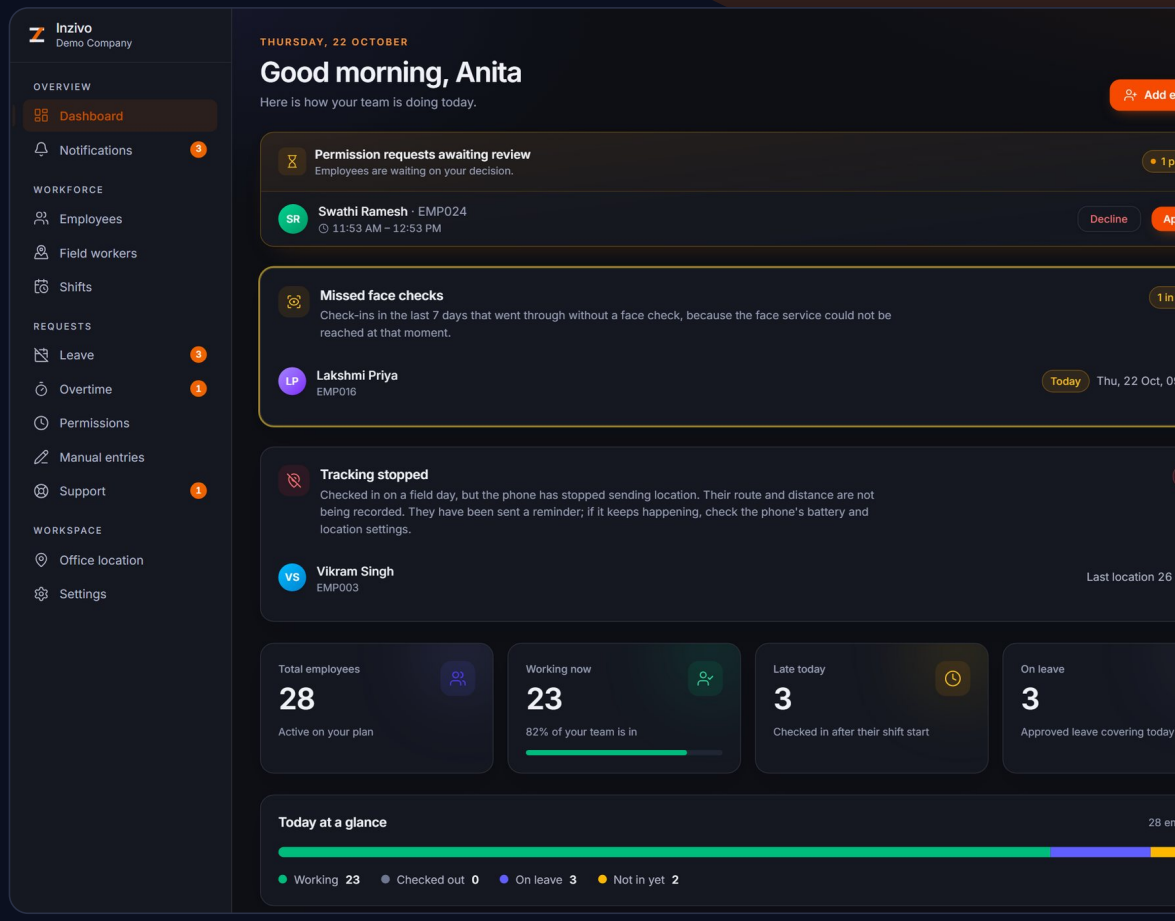
INZIVO

PROJECT PORTFOLIO

Attendance and field-force management platform

One platform for office, work-from-home and field teams: check-in with face verification, live route tracking, and everything an administrator needs in a single console.



Prepared by the Inzivo team

October 2026

OVERVIEW

One platform for every kind of working day

Inzivo is a multi-company attendance and field-force platform. Each company gets its own secure workspace: employees check in from an Android app or a shared kiosk, and administrators manage everything from a web console.

It serves the three kinds of working day with one system: people in the office, people working from home, and people out in the field. Each gets the right rules, checks and reports, and the owner sees a single, live picture of the team.

Admin web console

Dashboard, people, shifts, requests, alerts, field maps and reports.

Employee Android app

Check-in and out, requests, live tracking and in-app navigation.

Employee web app (My Page)

Employees check their status, hours and attendance calendar from any browser.

Also part of the platform: a shared kiosk check-in screen for each company, and a public API with keys for other systems.

3
work modes
Office, work from home and field

Face + PIN
verified check-in
Photo, PIN and face match, with flagging

Live
field tracking
Routes, stops and distance in the background

5 min
automation cycle
Reminders, alerts and automatic check-out

THE BUSINESS CASE

The problem, and what changes

The problem

- Attendance lives in registers, spreadsheets and chat photos, so it is slow to collect and easy to dispute.
- Nobody can prove who actually checked in, or that they were where they said.
- Field teams are invisible once they leave the office: no route, no record of visits, no real distance.
- Hours get argued over when a check-out is forgotten or a phone dies.
- Travel claims rest on guesswork, and reports take days to assemble.
- Office, remote and field staff end up on different tools with different rules.

WHAT INZIVO CHANGES

1 Attendance you can trust

Photo, PIN, face match and location are checked at every check-in.

2 A live picture

See who is in, late, on leave or out in the field, as it happens.

3 Fair, automatic hours

Reminders, automatic check-out and approved pauses keep hours accurate without chasing people.

4 Field work on the record

Routes, visits, photos and distance are captured as the day unfolds.

5 Reimbursement from real distance

Odometer and GPS distance feed travel claims, with an estimate when a reading is missing.

6 One system for every worker

Office, work-from-home and field staff share one rulebook, one console and one set of reports.

APPLICATION COMPONENTS

Three ways in, one system

Admin Dashboard

WEB CONSOLE

WHO USES IT

Owners, HR and managers

WHERE IT RUNS

Any desktop browser

WHAT IT DOES

- Live team dashboard and alerts
- People, shifts, offices and settings
- Approve leave, overtime and permissions
- Manual entries for missed days
- Field maps, routes and assigned places
- Excel and CSV reports, API keys

Employee Web App

MY PAGE

WHO USES IT

Every employee

WHERE IT RUNS

Any browser, same sign-in as the app

WHAT IT DOES

- Today's status and time so far
- Hours over the last 7 days
- Late marks this month and average start time
- Recent activity and an attendance calendar
- Change PIN and sign out
- View-only: check in with the app or kiosk

Mobile App

ANDROID

WHO USES IT

Employees, especially field staff

WHERE IT RUNS

Android phones

WHAT IT DOES

- Check in and out with selfie, PIN and face match
- Leave, overtime and permission requests
- Background tracking and live working time
- Today's places with turn-by-turn navigation
- Visit log with photo, contact and remarks
- Reminders and alerts

All three talk to the same backend and database, so what an employee does is visible to the admin straight away. Also part of the platform: a shared kiosk check-in screen for each company, and a public API with keys for other systems.

AUDIENCE

Who Inzivo is for

Inzivo is designed for companies whose people do not all sit at one desk: sales and service teams on the road, staff at customer sites, and offices that mix on-site and remote work.

Built for India: IST-based days, rupee reimbursement and your own public-holiday calendar.

Sales and distribution

FMCG, pharma and consumer-goods reps who visit dozens of shops a day: planned visits, guided routes and a record of every call.

Field service and maintenance

Installation, repair and utility teams: assign the day's jobs, see arrival times, and measure time and distance.

Collections, recovery and surveys

Banks, NBFCs and research agencies: every visit is stamped with place, time, photo and contact notes.

Logistics and delivery

Know how far each person travelled, with distance and reimbursement worked out from the odometer or GPS.

Contract and customer-site staff

People who work at a client's premises check in with a photo, PIN and face match, so attendance can be trusted.

Hybrid offices

Office staff are checked against the office area and remote staff against their home location, under one rulebook.

WHO USES IT DAY TO DAY

Owners and HR

See who is in, late or on leave, approve requests and export reports.

Operations managers

Watch field teams, assign places, and review routes and visits.

Finance

Reimbursement worked out from distance, plus Excel and CSV exports.

Employees

A simple phone app: check in, make requests and follow the day's route.

INZIVO IN NUMBERS

The product at a glance

3
work modes
Office, work from home and field, each with its own rules

5 min
automation cycle
Reminders, alerts and automatic check-out run every five minutes

15 s
to see new requests
New requests and alerts appear live in the admin console

15 min
after shift end
A forgotten check-out is closed automatically

98
API endpoints
Behind the admin site, the app, the kiosk and partner access

12
admin screens
Dashboard, people, requests, field force and settings

15
app screens
In the employee Android app, from check-in to navigation

31
data tables
One database holding every company's data, kept separate

10 s
GPS sampling
For field workers, sent in batches every two minutes

< 1 MB
mobile data a day
For location tracking over a full eight-hour field day

500
people per upload
Add a whole team from one spreadsheet

45 days
photo retention
Check-in photos are then removed automatically

Figures describe the product as built in October 2026. Data use comes from our own measurement of an eight-hour field day.

CAPABILITIES

What each group of people gets

Administrators

- **Live dashboard:** who is working, late, on leave or not in yet
- **Add people** one by one, or up to 500 at once from a spreadsheet
- **Shifts** per weekday, with office and home locations and geofence radius
- **Approve** leave, overtime and short permissions in one tap
- **Manual entries:** record a day the phone missed, with a reason and an audit trail
- **Alerts centre** for stopped tracking and missed face checks
- **Excel and CSV reports**, plus a public API with keys

Employees

- **Check in and out** with a photo, PIN and face check
- **See** working time, leave balance, history and weekly hours
- **Request** leave, overtime or a short permission, and report an issue with a screenshot
- **Reminders** before the shift and when a check-in is missed
- **Automatic check-out** if they forget at the end of the shift
- **My Page on the web:** status, weekly hours, late marks and a calendar
- **Biometric unlock,** profile photo and change-PIN in the app

Field teams

- **Background tracking** of the route, distance and places visited
- **Log each visit** with photo, contact person and remarks
- **Assigned places** in nearest-first order, with turn-by-turn directions in the app
- **Odometer** start and end, and reimbursement worked out from distance
- **Nearby** shops, schools and restaurants on the map
- **Automatic switch** between Field and Office when they reach the office

Role-based by design

Each person sees only the tools their role needs, and each company's data is kept separate from every other company's.

THE EMPLOYEE APP

Four tabs, one simple routine

ANDROID APP · REACT NATIVE WITH EXPO

SIGN IN

Organisation code → Employee code and PIN → First time: face enrolment → Optional fingerprint unlock

Dashboard

Live map · Today's places · Navigate

- Live working time and distance to the office
- Leave balance, recent history and weekly hours
- Late and leave notices
- Field workers: today's places and in-app navigation

Check In/Out

Mode · Selfie · PIN · Slide to confirm

- Pick today's mode: Office or Field (Home or Office for remote staff)
- Take a selfie, enter the six-box PIN and slide to confirm
- Field workers enter start and end km
- A clear message if something blocks the check-in

Requests

Leave · Overtime · Permission

- Leave: casual, sick or earned
- Overtime: set an end time, then add a summary and photo
- Permission: a short, timed pause
- See the status of every request

Profile

History · Change PIN · Report an issue

- Attendance history
- Change PIN and profile photo
- Report an issue, with an optional screenshot
- Fingerprint unlock and sign out

BEHIND THE SCENES

Background tracking

Location every two minutes while checked in, with ten-second samples for field workers.

Guided phone settings

Checks 'Allow all the time' and battery saver, and shows a Fix now prompt if tracking is at risk.

Push notifications

Shift reminders, leaving-the-area warnings, tracking alerts and automatic check-out notices.

THE EMPLOYEE APP

A check-in, step by step

1 Choose the mode

Office or Field. Remote staff choose Home or Office.

2 Take a selfie

A photo is required. It is shrunk on the phone so it uploads quickly.

3 Enter the PIN

Six boxes, then slide to confirm.

4 Send

Photo, PIN and location go to the server. Field workers add their start km.

5 Checks run

A handful of quick checks decide whether to accept it (below).

6 Done

Checked in, marked on time or late against the shift, and tracking starts.

What the server checks

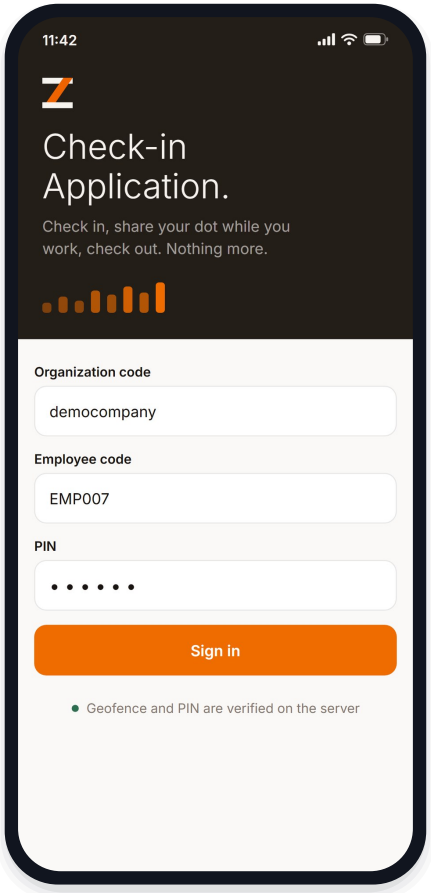
- The PIN is correct, and there have not been too many wrong tries
- The face matches the one enrolled. If the face service is down, the check-in is accepted and flagged for review
- Office or home days: inside the allowed area. Field days: no area limit
- The location is real: mock-GPS apps are rejected
- Not on approved leave today
- Only one check-in and one check-out per day

Then, in the background

- Location is sent every two minutes. Field workers are also sampled every ten seconds
- Leave the work area and a warning arrives; after ten minutes working time pauses
- With no signal, samples are kept on the phone and sent later
- The app checks phone settings and shows a Fix now prompt if tracking is at risk

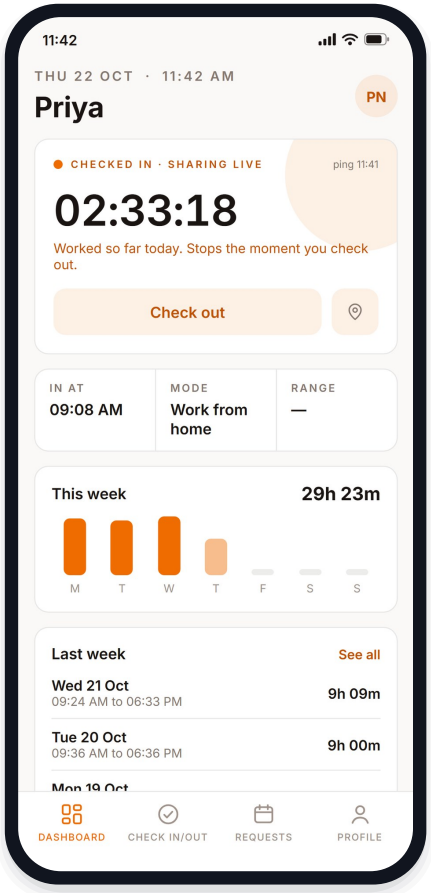
THE EMPLOYEE APP

The app, screen by screen (1)



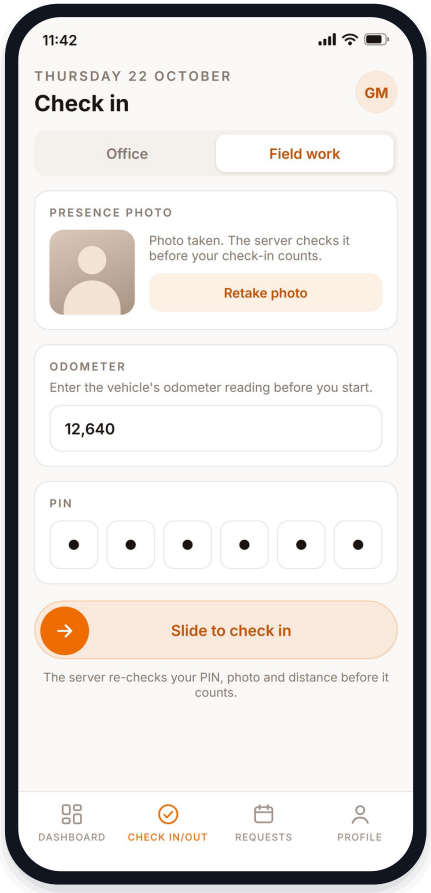
Sign in

Organisation code, employee code and PIN



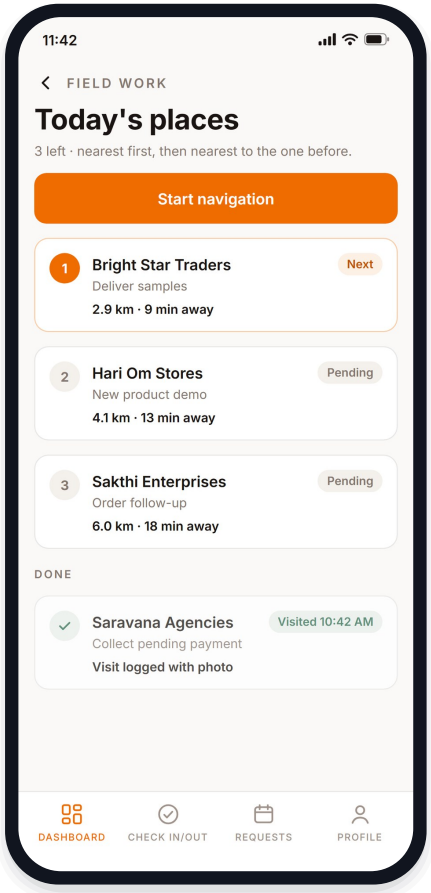
Dashboard

Live working time, leave balance and history



Check in and out

Mode, selfie, PIN and slide to confirm



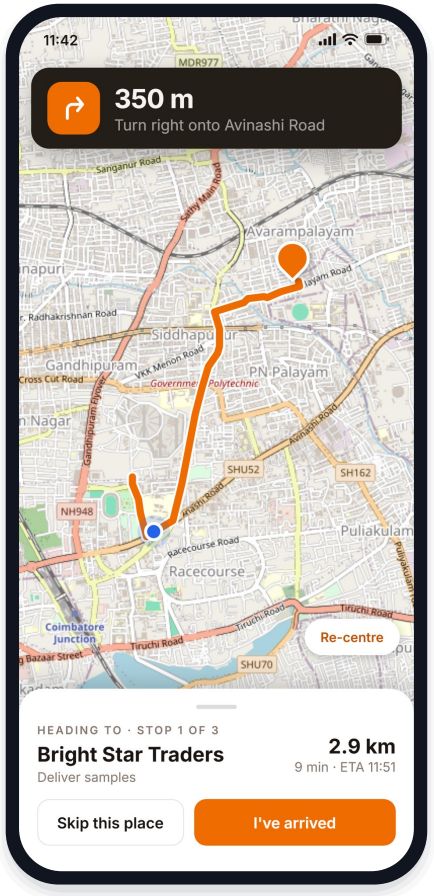
Today's places

The day's stops, nearest first

Illustrative screens: drawn from the app's design with sample data, to show each screen. They are not live phone screenshots.

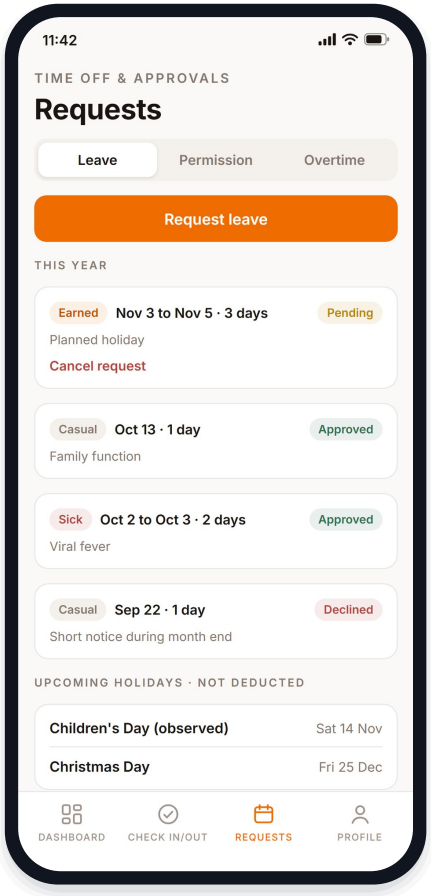
THE EMPLOYEE APP

The app, screen by screen (2)



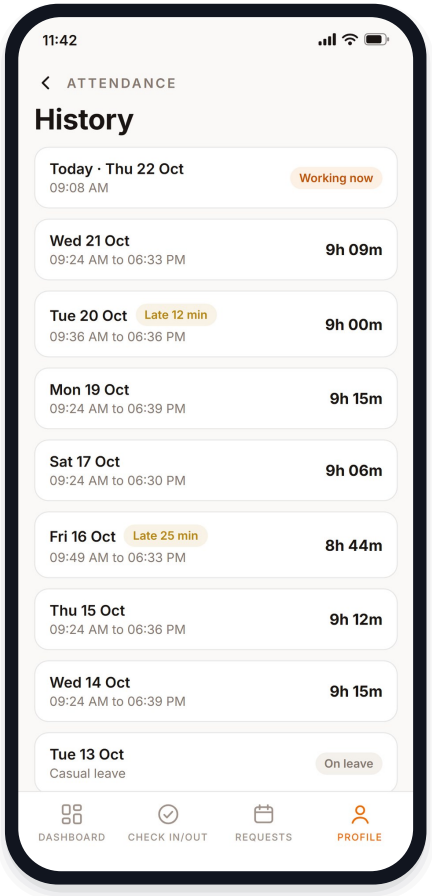
Navigation

Turn-by-turn directions inside the app



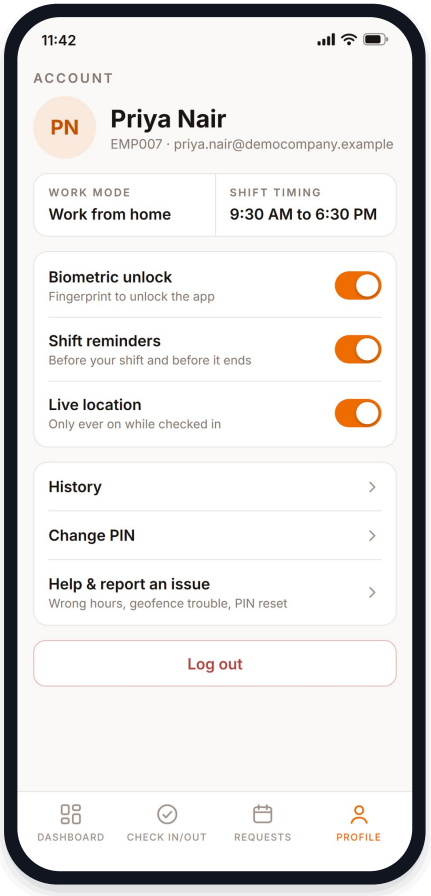
Requests

Leave, overtime and permission



Attendance history

Every day at a glance



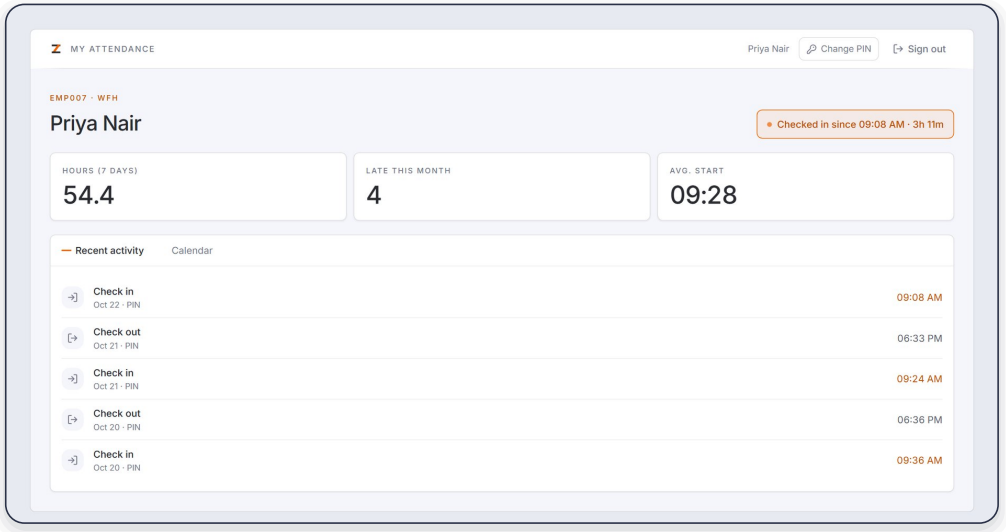
Profile

PIN, photo and settings

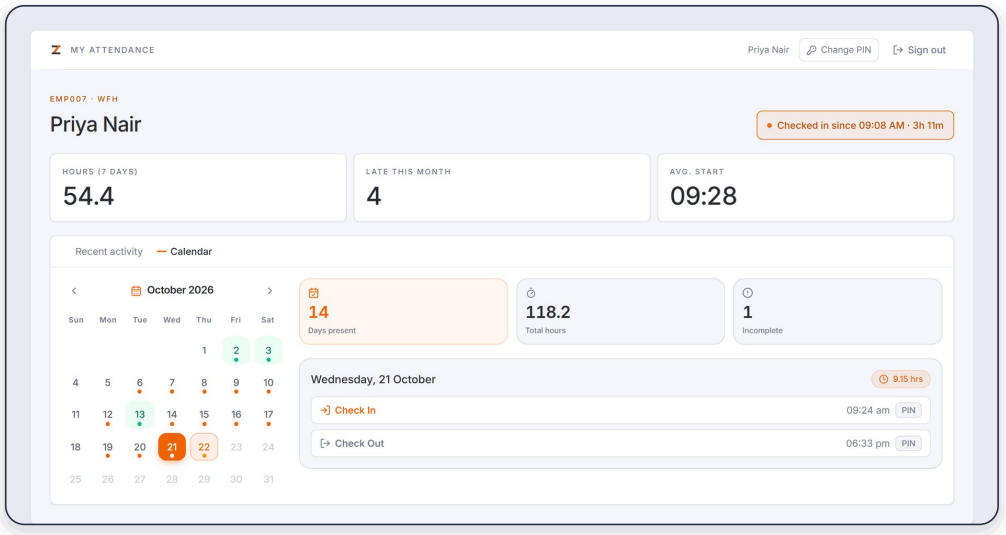
Illustrative screens: drawn from the app's design with sample data, to show each screen. They are not live phone screenshots.

THE EMPLOYEE WEB APP

My Page: the employee's own view



Recent activity: every check-in and check-out, newest first, with the hours so far today.



Calendar: days present, total hours and any incomplete day, with the times of the selected day. Leave days are shaded.

What an employee sees

- **Today's status** at a glance, such as checked in since 9:08 AM with the time worked so far
- **Hours over the last 7 days,** late marks this month and the average start time
- **Recent activity:** each check-in and check-out with its time and how it was made
- **A monthly calendar** with days present, total hours and incomplete days. Select a day for its times
- **Change PIN** and sign out. Sign-in is the same organisation code, employee ID and PIN as the app

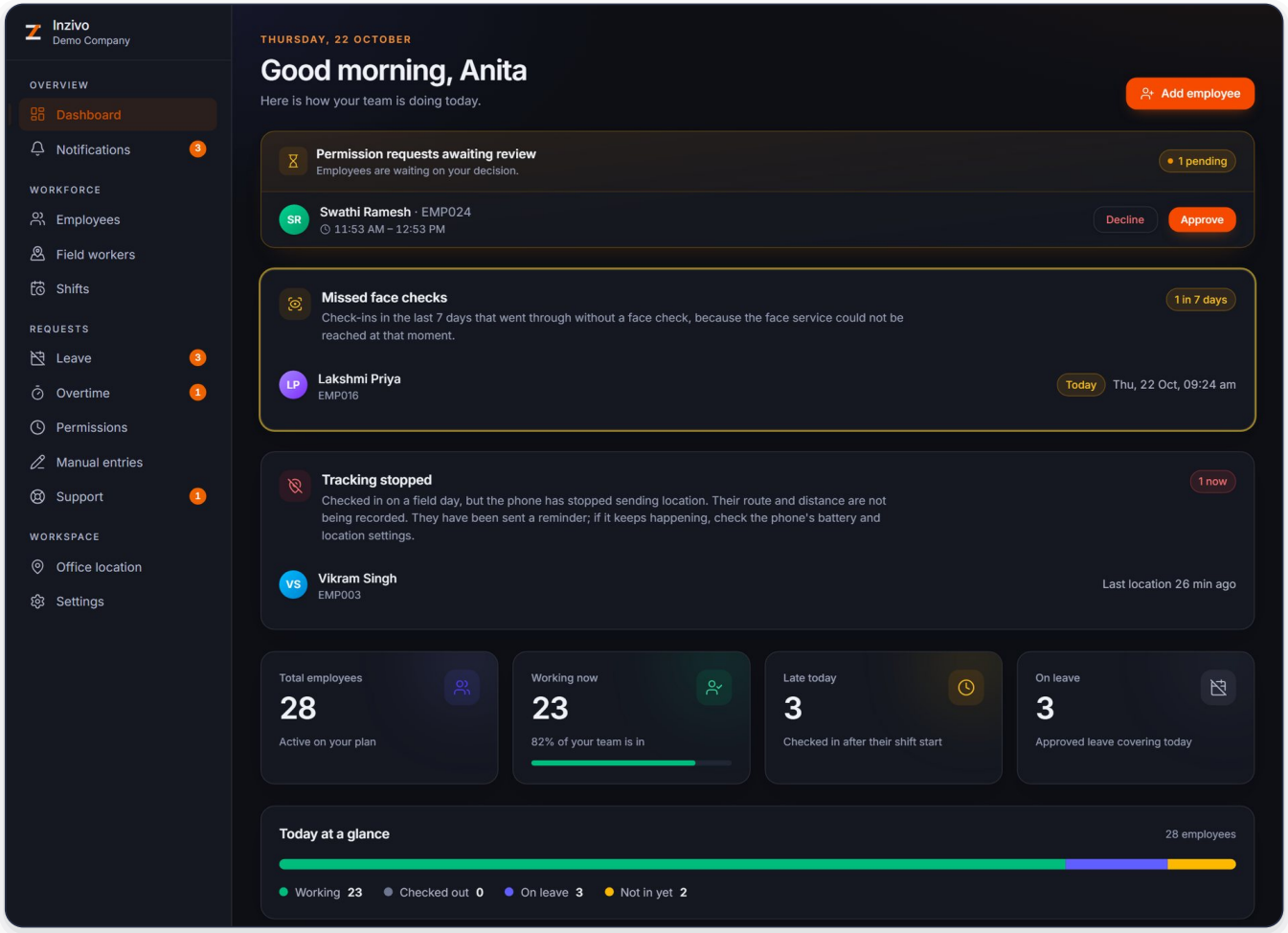
View-only by design

Employees check in with the Android app or the kiosk. My Page lets them review their own record from any laptop or phone browser.

Actual screenshots of the employee web app, with sample data.

ADMIN CONSOLE

A dashboard that shows what needs attention



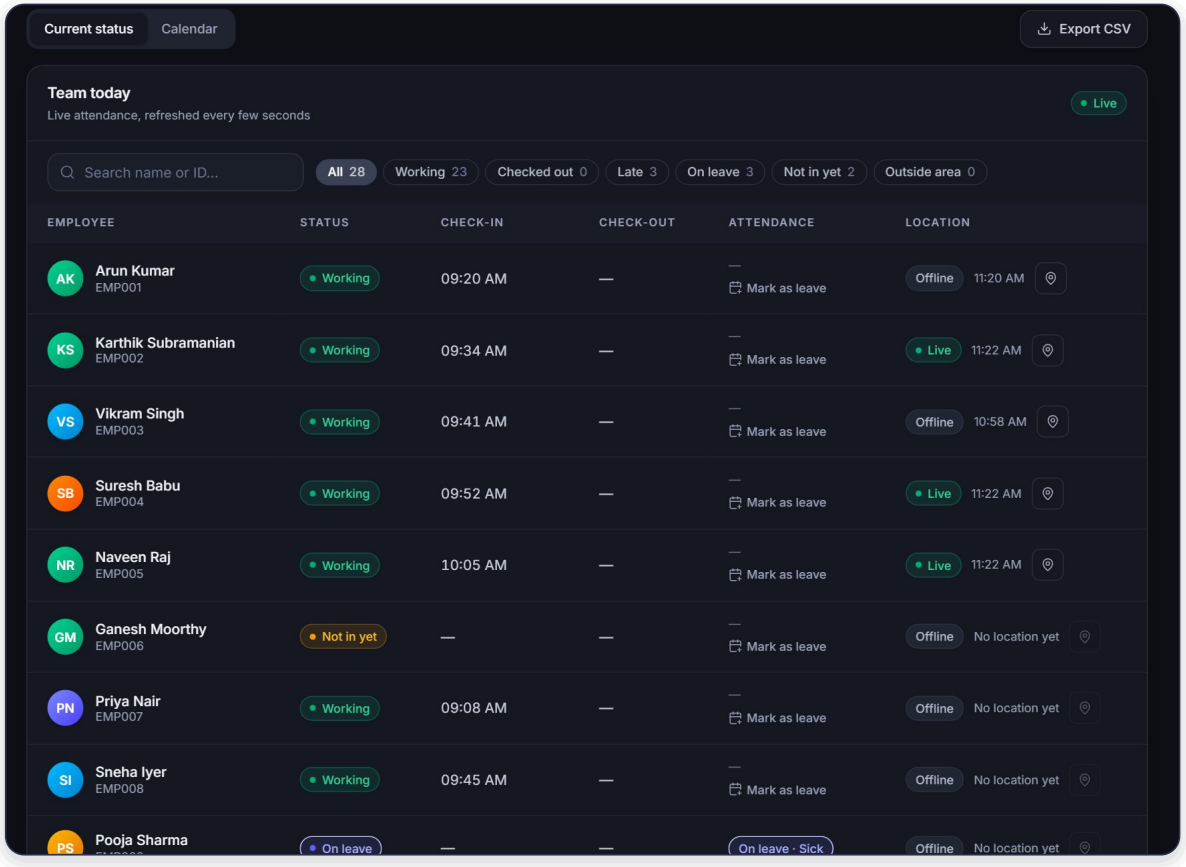
Today's picture: alerts that need a decision, live counts, and the attendance mix, on one screen.

What it shows

- Everything that needs attention sits at the top: permission requests, missed face checks and field workers whose tracking stopped.
- Live counts of total employees, working now, late today and on leave.
- A **Today at a glance** bar and the **Team today** table, with filters, search and CSV export.
- New requests and alerts **appear on their own** within about 15 seconds, with no page refresh.
- Light and dark themes, and a guided tour on first use.

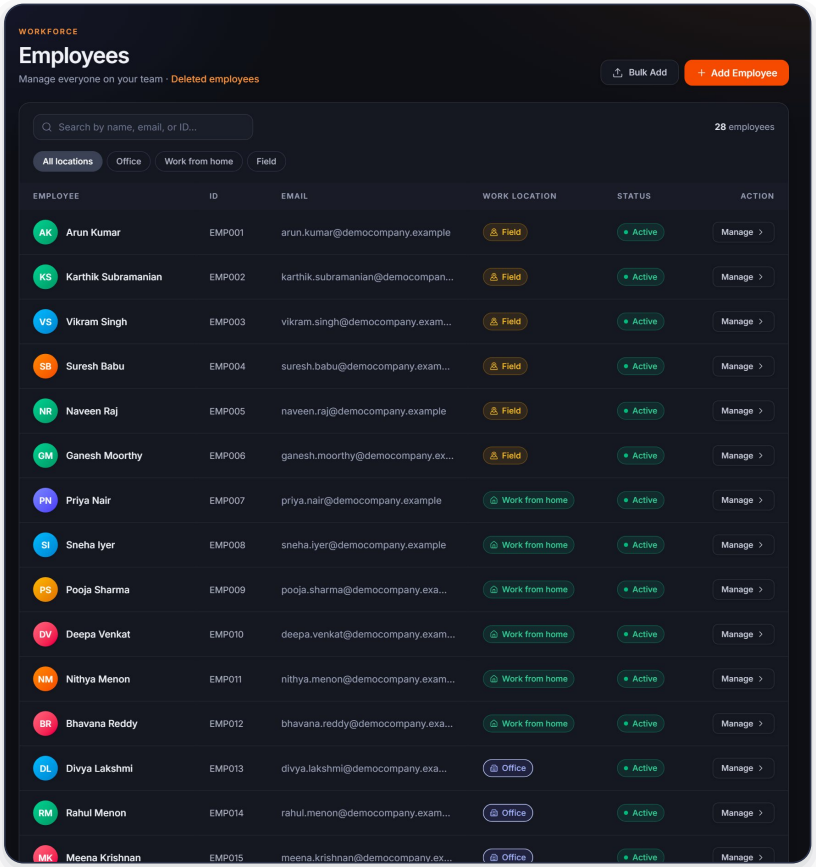
ADMIN CONSOLE

Managing the team day to day



Team today: live status, check-in times, leave marks and the last known location of each person.

- Filter by working, checked out, late, on leave, not in yet or outside the area
- Mark someone on leave straight from the table
- Face-check badges and photos on every attendance record

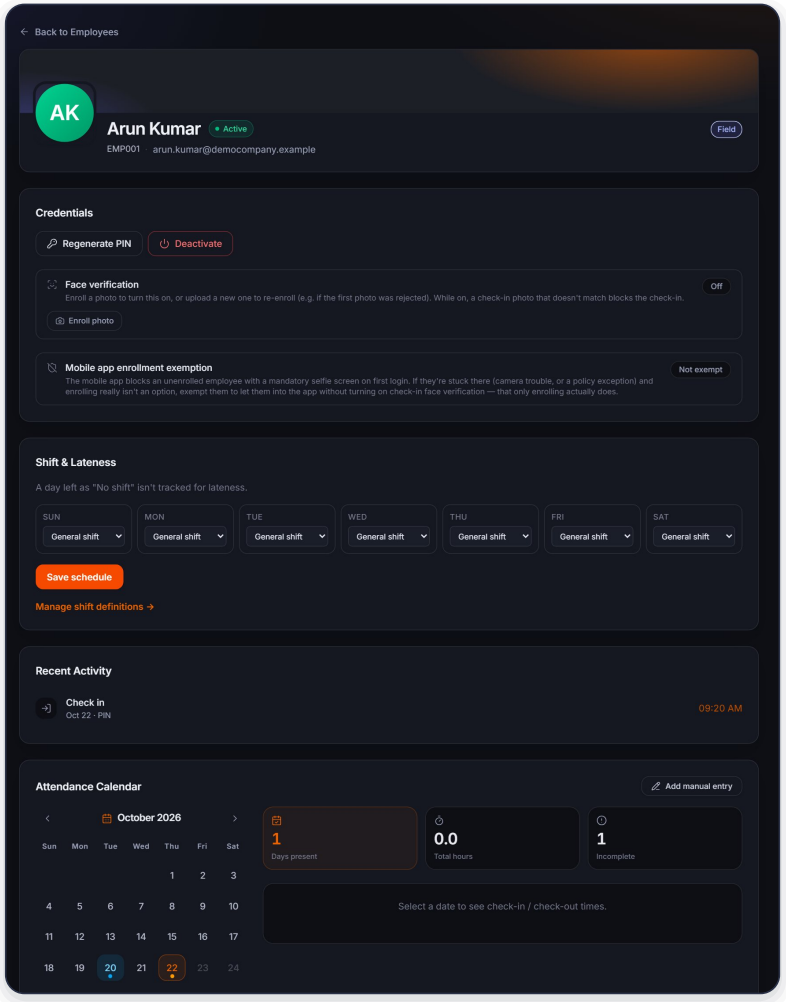


Employees: add one person or up to 500 from a CSV, with auto-generated IDs and PINs.

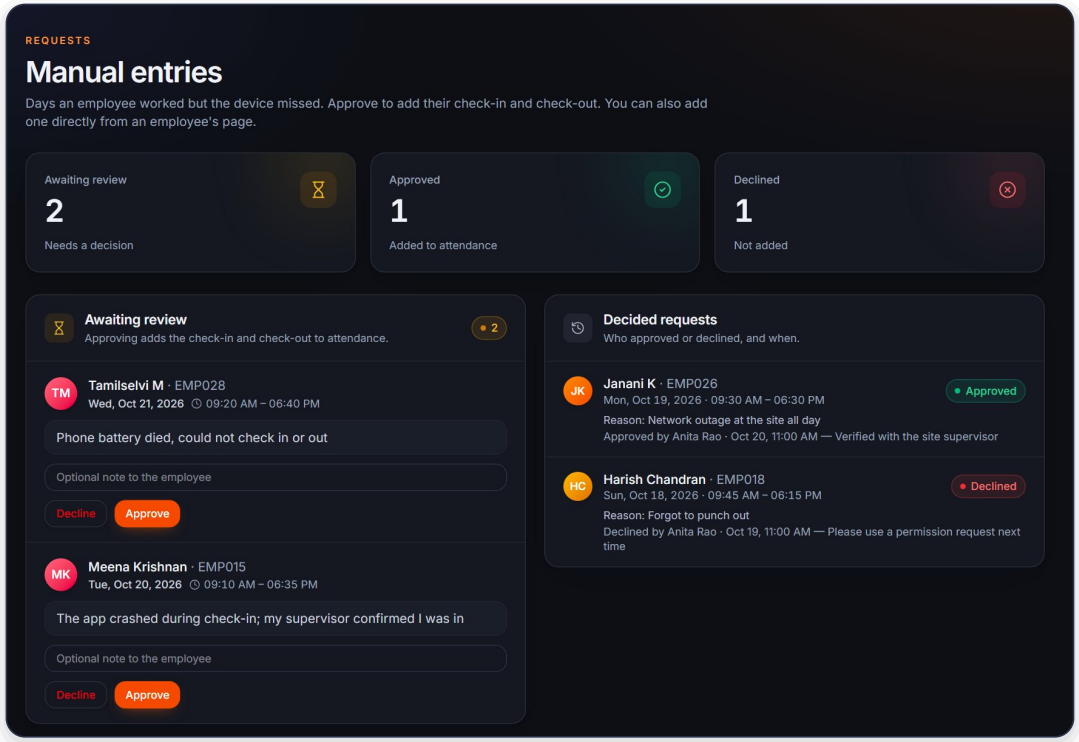
- Three work modes: Office, Work from home, Field
- PIN reset, deactivate and archive

ADMIN CONSOLE

Each person's record, and fixing a missed day



An employee's page: credentials, face-check settings, shifts for each weekday and an attendance calendar.

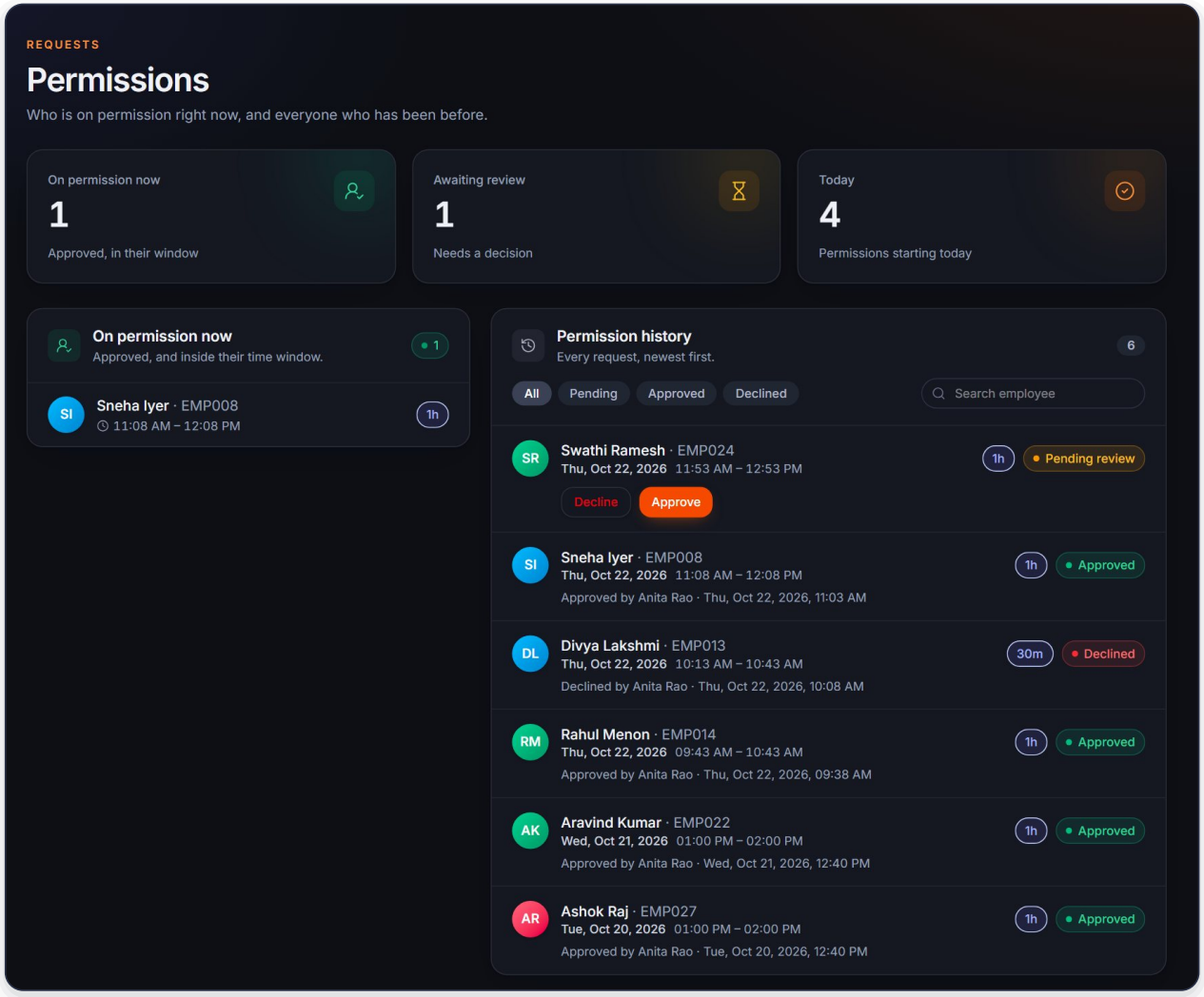


Manual entries: days an employee worked but the device missed, waiting for an admin decision.

- An employee can ask for a missed day to be added, with a reason. An admin can also add one straight from the employee's page
- Approving adds the check-in and check-out as a **manual** record, with the reason and the approving admin's name
- Entries can go back seven days and cannot be in the future. Lateness is worked out against that day's shift, like a real punch
- A day that already has a record cannot be duplicated

ADMIN CONSOLE

Permissions: who is paused, and the full history



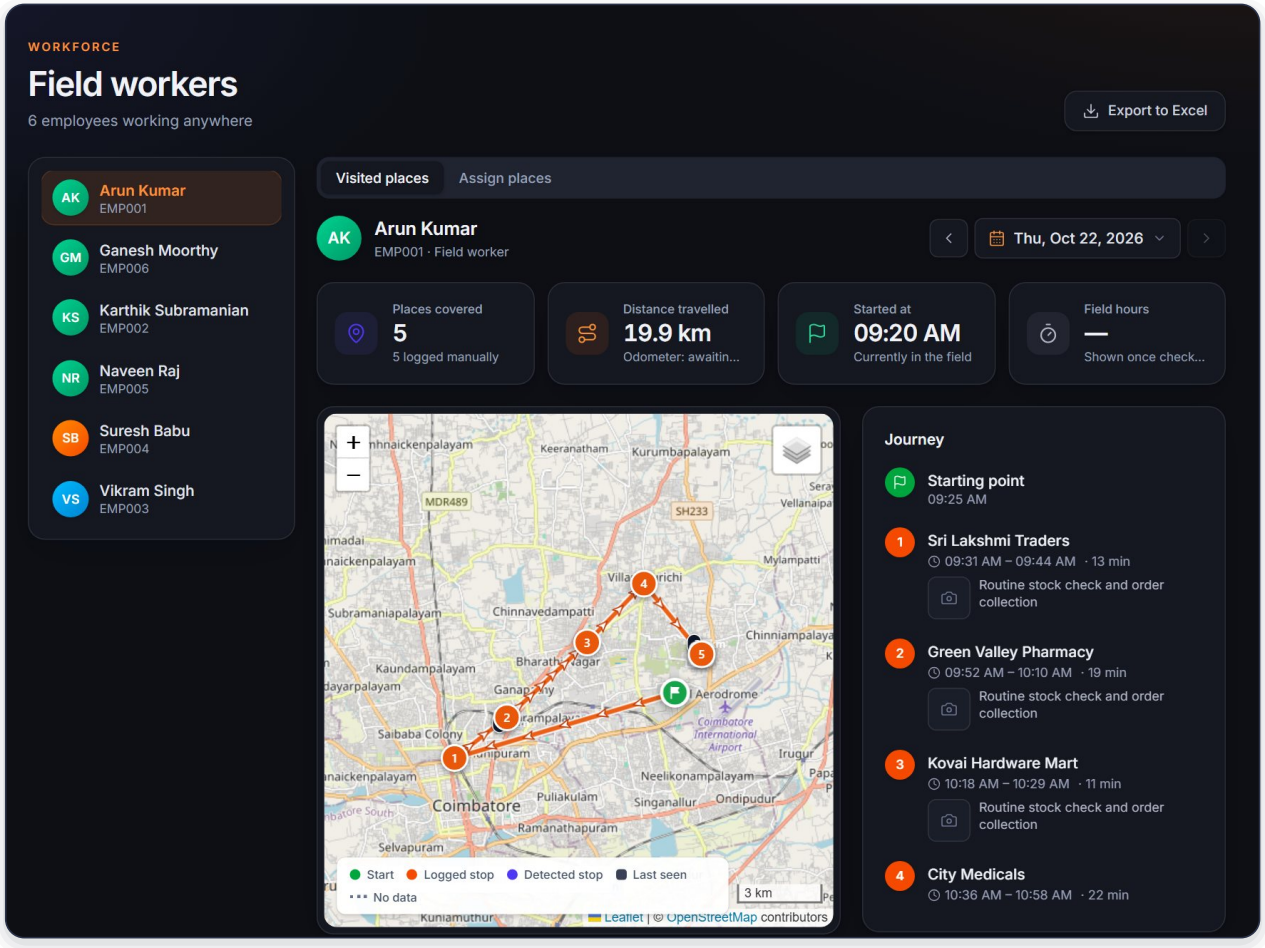
Permissions: who is on a permission right now, and every request with its decision and who made it.

How a permission works

- An employee asks for a **short, timed pause** in the app, for example an hour for an errand
- The admin approves or declines from the dashboard, Notifications or this page. Nothing changes for the employee until then
- Once approved, working time is paused for that window and resumes by itself
- Everything is kept: who asked, when, the window, who decided and when. Search and filter the history
- While someone is on an approved permission, automatic check-out leaves them alone

FIELD FORCE

Every field day, on one screen



Pick any worker and date: places covered, distance, start time and the full route with numbered stops.

What the admin sees

- **Route map** with numbered stops, direction arrows and Street or Satellite layers
- **Journey list** with arrival, departure and time spent at each place
- **Visit details** logged by the employee: contact person, phone, remarks and site photos
- **Tracking gaps** are marked and left out of the distance, so numbers stay honest
- **Odometer and reimbursement:** start and end km, GPS versus odometer, and a reimbursement form
- **One-click Excel report** for several employees and up to 31 days

FIELD FORCE

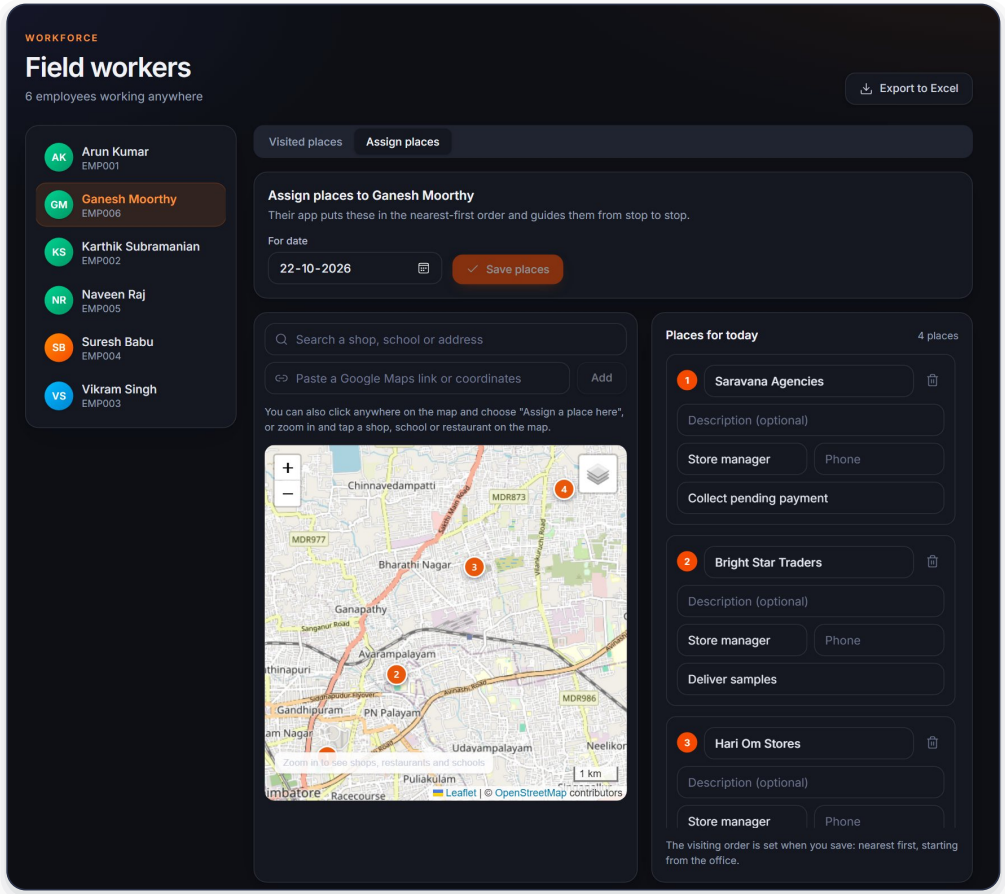
Assign the day's places, and guide the way

From plan to visit

- Add places by **searching**, **clicking the map**, or **pasting a Google Maps link** or coordinates
- The employee's app puts them in **nearest-first order** and re-orders after every stop
- Each stop opens the visit form with the details already filled in; a stop can be skipped with a reason
- Visited places appear in the day's journey with times, details and photos

Navigation inside the app

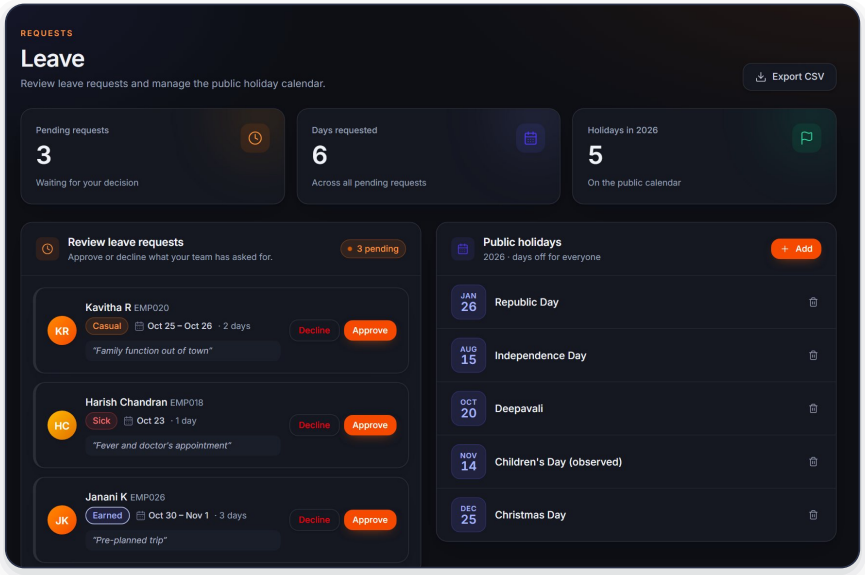
Turn-by-turn directions with spoken prompts, automatic re-routing and arrival detection, so field workers never leave the app. Built on OpenStreetMap data and OpenRouteService, with no Google Maps account needed.



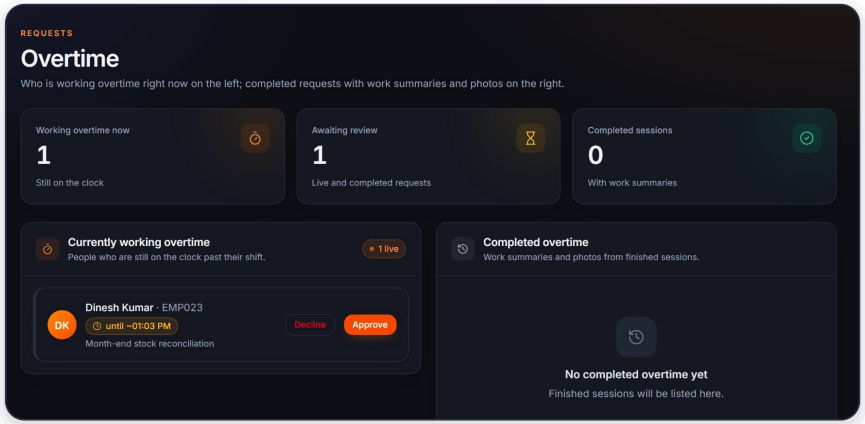
Assigning places to a field worker: search, click the map, or paste a map link.

ADMIN CONSOLE

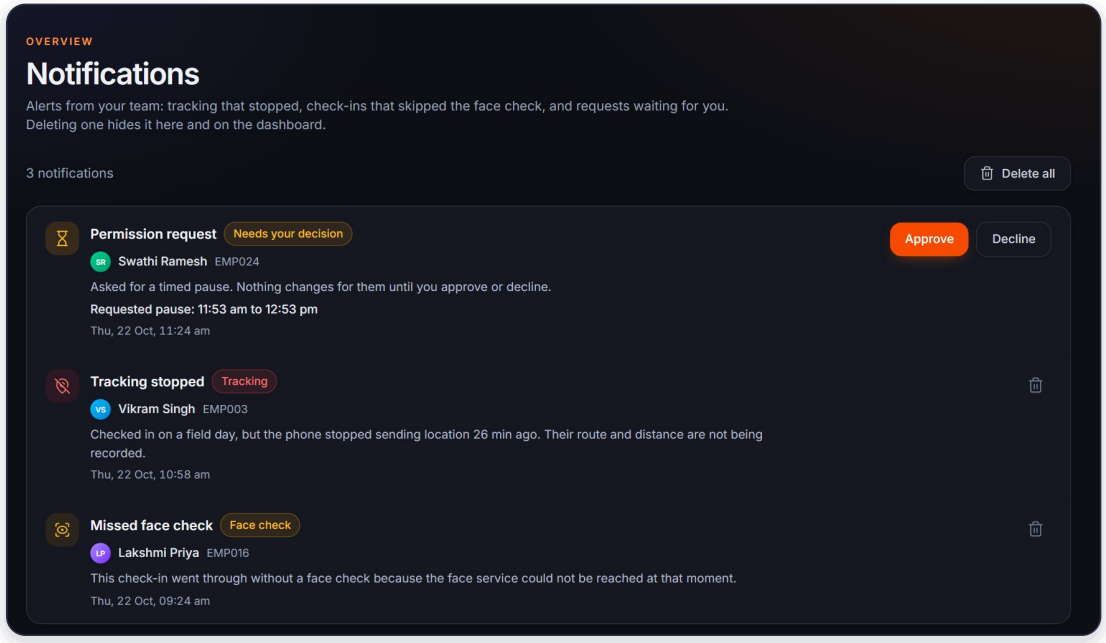
Requests and alerts in one place



Leave: approve or decline, with a public-holiday calendar and CSV export.



Overtime: who is working past shift right now, and the summaries and photos they submit.

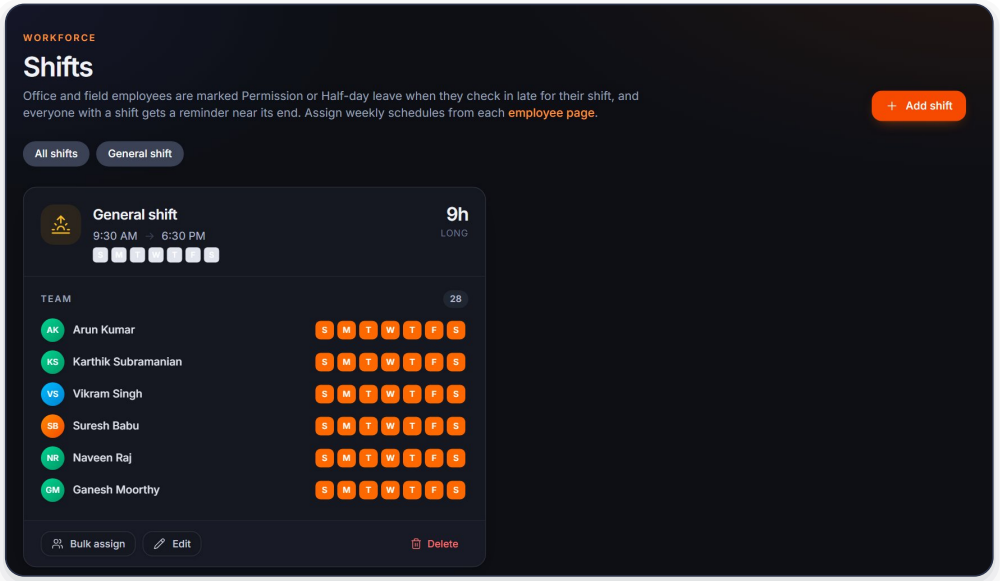


Notifications centre: every alert in one list, each with a delete button.

- Permission requests are approved or declined right here
- Stopped tracking and missed face checks are listed with the person and the time
- The sidebar badges and lists **update live** without reloading

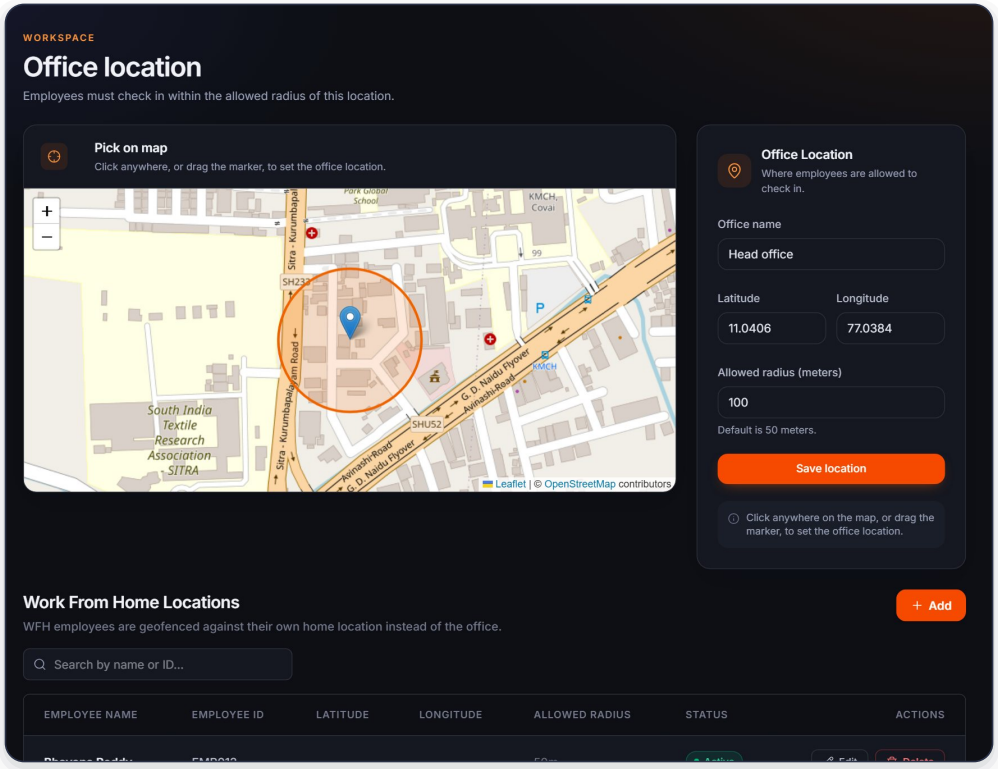
ADMIN CONSOLE

Set it up once: shifts, offices and rules



Shifts: each shift runs on the weekdays you choose, and people are assigned for each day or in bulk.

- Any number of shifts; lateness is measured against the shift
- Settings for photo on check-out, reimbursement rate, leave quotas and the lateness cutoff



Office location: the check-in area is a circle on the map; work-from-home staff get their own home location.

- Geofence radius set per office and per home location
- Administrators and API keys managed in Settings

ORGANISATION SETTINGS

Photo on check-out

Require a presence photo to check out, the same as at check-in.

Reimbursement rate

A default rate per km for field workers, adjustable per person and day.

Leave quotas

Days per year for casual, sick and earned leave.

Lateness cutoff

How late a check-in can be before it is marked on the record.

HOW IT WORKS

A working day for each kind of employee

OFFICE

1. Open the app
2. Photo, PIN and face match
3. Inside the office area?
4. Work. Time pauses if they leave the area
5. Check out

WORK FROM HOME

1. Choose Home or Office
2. Photo, PIN and face match
3. Inside the home area?
4. Work
5. Check out

FIELD

1. Check in, no area limit
2. Today's places, nearest first
3. Navigate and log visits
4. Route and km are recorded
5. Check out with end km

The same rules apply to everyone

- One check-in and one check-out a day, enforced by the database
- Lateness is measured against the person's shift for that weekday
- A forgotten check-out is closed automatically, so hours never stay open overnight
- Approved leave blocks check-in; public holidays switch reminders off
- Face and location checks happen at every check-in
- Admins can correct a missed day with a manual entry

AUTOMATION

What happens automatically: before and during the shift

BEFORE THE SHIFT

Checked every five minutes, for everyone with a shift that day

1. **10 minutes before the shift**
   A push says "Your shift starts soon" with the start time. Sent once.

2. **5 minutes after the shift starts**
   If the person still has not checked in, a push says "Time to check in". Sent once, and dropped if it would arrive more than an hour late.

Nobody is reminded who has already checked in, is on approved leave, or whose company has a public holiday that day. People with no shift entered for that weekday are skipped.

DURING THE SHIFT

Driven by the phone's location, about every two minutes

3. **Leaving the work area**
   After three readings in a row outside the area, a push warns "Return within 10 minutes or your working time will be paused". If they do not return, time is paused; when they are back inside it resumes.

4. **Field worker reaches the office**
   After three readings in a row at the office, the day switches from Field to Office by itself, so indoor time is not counted as travel. Leaving switches it back.

5. **Phone stops sending location**
   After ten minutes of silence, the worker gets a push to fix location sharing and the admin sees a Tracking stopped alert. Re-sent every 30 minutes for up to six hours.

6. **Shift ends in 10 minutes**
   A push says "Your shift is ending soon" (or "Your overtime is ending soon" during approved overtime). Sent once.

Timings are checked every five minutes, so a push can arrive up to five minutes after its exact time.

AUTOMATION

What happens automatically: at the end of the day

AFTER THE SHIFT

7 — 10 minutes after the shift ends

Still checked in? A push says "Your shift has ended. Please check out now, or you will be checked out automatically in 5 minutes." (During approved overtime, the clock starts from the overtime end time.)

8 — 5 minutes later, with no response

The check-out is recorded **at the shift end time**, so working hours stop where the shift did. Open pauses, the Field/Office timeline and any open overtime request are closed, and the person gets a notice.

9 — Field workers: end km is filled in

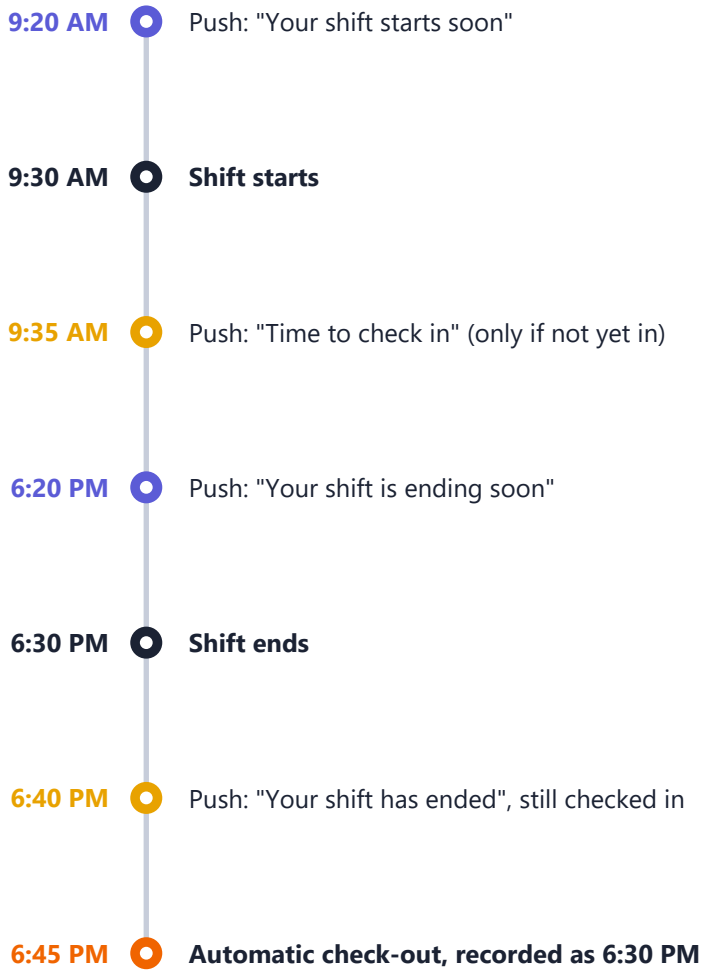
If a start km was entered, the end km is the start plus the GPS distance travelled in Field mode, marked as an estimate, so the reimbursement can still be worked out.

10 — A forgotten check-out from an earlier day

The next time that person checks in, any earlier open check-in is closed automatically at that day's shift end (or the end of the day).

Left alone: people on an approved permission, people with no shift entered for that day, and anyone who checked in after the shift end.

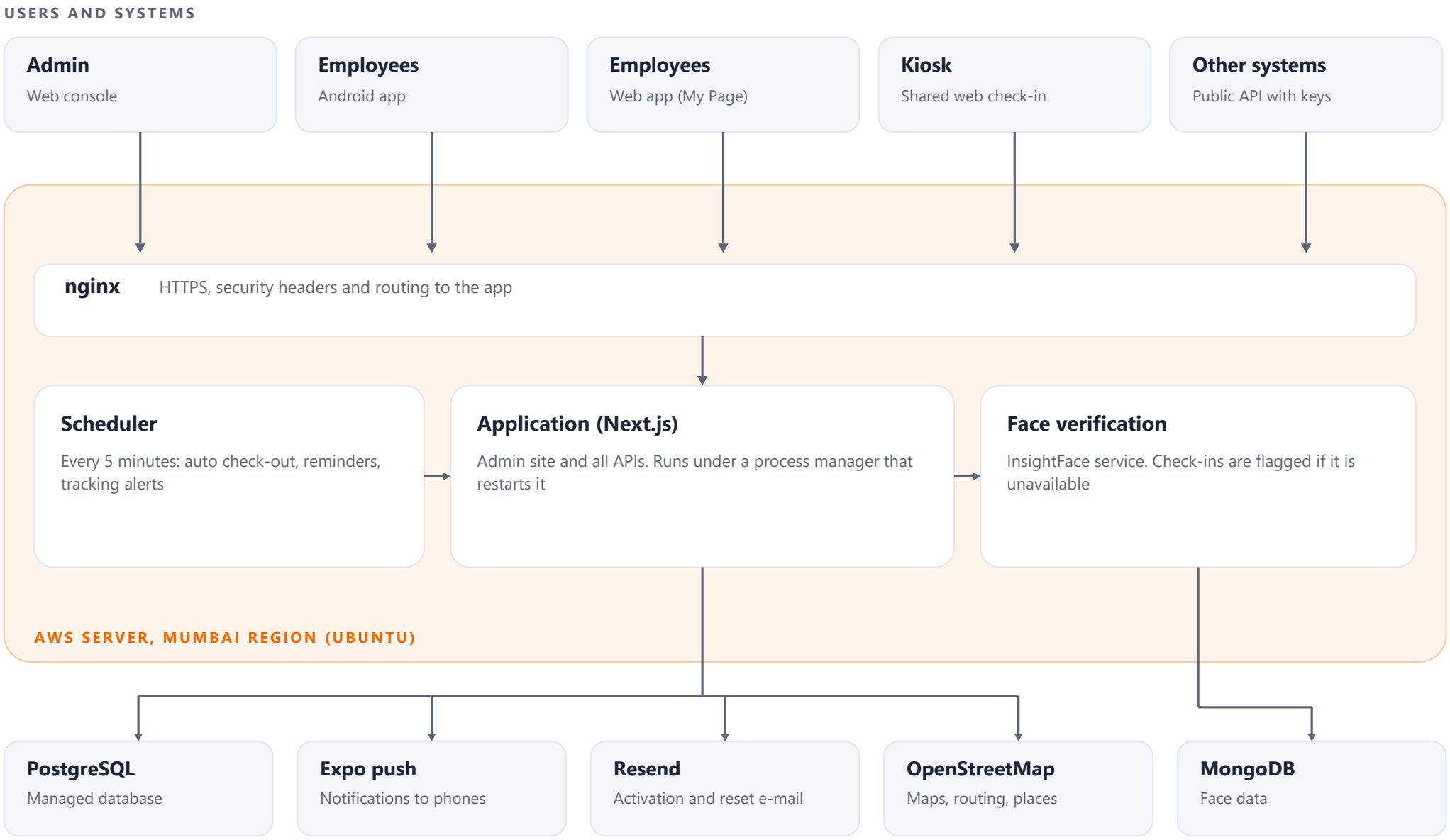
EXAMPLE: A 9:30 AM TO 6:30 PM SHIFT



Times are targets: the scheduler runs every five minutes, so each push or check-out lands within five minutes after its target. Every automatic action is recorded, and an admin can correct it with a manual entry.

ARCHITECTURE

How the system fits together



SECURITY AND RELIABILITY

Built to protect people's data and keep running

Security

- **Secure connections:** HTTPS throughout, strict transport security, a content-security policy, clickjacking protection and no-sniff headers
- **Credentials:** passwords and PINs are stored only as bcrypt hashes; chosen PINs must be at least 6 digits
- **Brute-force protection:** per-person attempt limits (10 per 5 minutes, 30 per hour) shared by kiosk, app and web, plus per-address limits on sign-in, kiosk, password reset and other public endpoints
- **Sessions:** separate signing keys for web and mobile, 15-minute access tokens, refresh tokens that rotate and can be revoked, and sign-out everywhere in one step
- **Company isolation:** data access is scoped to the signed-in company, keeping each company's data separate
- **Presence integrity:** face verification, geofence checks, mock-GPS rejection, and one check-in and check-out a day enforced by the database
- **Audit trail:** manual entries and permission decisions record who approved them and why
- **Safe data handling:** every request is validated, database access is parameterised, photos are type-checked, size-limited, resized and removed after 45 days
- **Integrations:** API keys stored hashed, payment webhooks signature-checked, scheduler protected by a secret
- **Kept current:** dependencies audited and patched, with critical findings closed in October 2026

Reliability and operations

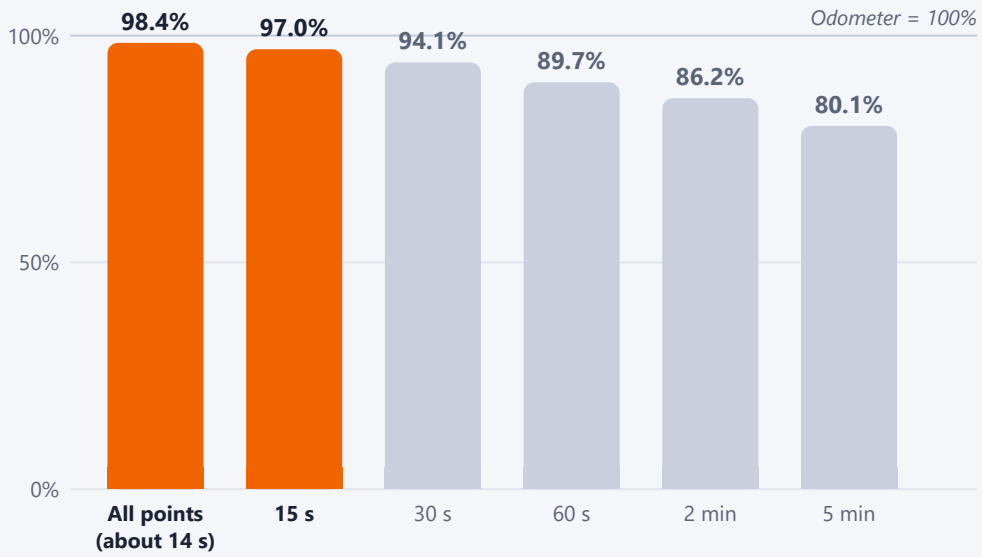
- **Hosting:** AWS in the Mumbai region behind nginx, with the app under a process manager that restarts it automatically
- **Database:** managed PostgreSQL with connection pooling and per-process connection limits
- **Face verification** runs on the same server as a boot-time service. If it is ever unavailable, check-ins continue and are flagged for review instead of blocking work
- **Health endpoint** reports database and face-service status for monitoring
- **Scheduler** runs every 5 minutes for auto check-out, shift reminders and tracking-stopped alerts
- **Mobile resilience:** unsent location points are kept on the phone and sent later, and the app guides battery and location settings so tracking survives
- **Live admin screens:** new requests and alerts appear within about 15 seconds

SECURITY AND TESTING

Measured, not assumed: GPS accuracy and quality checks

Distance captured, compared with the vehicle's odometer

One real 9-hour field trip, re-sampled at different intervals



Orange: the frequent sampling the app uses now. Grey: coarser intervals, including the old two-minute setting.

A measured trip

Vehicle odometer: 135 km. GPS distance with a point about every 14 seconds: 132.8 km, which is 98.4% of the odometer.

What the measurement showed

- Sending a point only every two minutes cut corners and under-counted this trip by 18.6 km (86%)
- Sampling every 10 to 15 seconds and sending the points in batches brought it to about 98%
- The last 1.6% is normal: odometers often read slightly high, GPS jitter is filtered out, and signals drop for a few minutes

Data and honesty

- Tracking costs under 1 MB of mobile data a day for a full eight-hour field day
- This is one measured trip. More trips are compared against the odometer as they complete

Other quality checks

Type-checked and linted code, a dependency audit with critical findings fixed, and end-to-end runs of the newest features on throwaway test companies.

TECHNOLOGY

What it is built with

Admin console and APIs	Next.js 16 and React, TypeScript, Tailwind CSS
Employee app	React Native with Expo (Android), cloud builds with EAS
Employee web app	Next.js and React, served from the same application as the admin console
Database	PostgreSQL 17 (managed), accessed through the Prisma ORM
Sign-in	NextAuth for administrators, signed tokens with refresh for the mobile app, PIN for kiosk
Face verification	InsightFace (ArcFace) service on FastAPI, with MongoDB for stored face data
Maps and routing	OpenStreetMap tiles, Leaflet, OpenRouteService for directions, Overpass for nearby places
Notifications and e-mail	Expo push notifications, Resend for transactional e-mail
Reports	Excel (ExcelJS) and CSV exports, public REST API with API keys
Hosting	AWS EC2 in Mumbai, Ubuntu, nginx, a process manager, a 5-minute scheduler

STATUS

Where things stand

Delivered and operating

- Admin web console, Android app, employee web app (My Page), shared kiosk and public API
- Office, work-from-home and field workflows
- Face-verified check-in and geofencing
- Live field tracking, visit logging and assigned places with in-app navigation
- Automatic check-out, reminders and alerts
- Excel and CSV reports
- Manual attendance entries with approval and an audit trail
- Permissions history and a live view of who is on permission
- Shift weekdays and bulk assignment
- Refreshed admin console design and kiosk screen
- Payment gateway for plan sign-up
- Verified e-mail sending domain
- Live on AWS with security hardening in place

In progress

CHECK-IN AND LOCATIONS

- Multiple office and site locations, each with its own check-in radius
- Offline check-in that syncs when the signal returns
- Live-face check to block photos of another person
- iPhone version of the employee app

FIELD WORK

- Forms and checklists to complete at each assigned place
- Expense claims with photos: tolls, parking, meals
- Live map of the whole field team at once

ADMIN AND INTEGRATIONS

- Payroll export of attendance, overtime and leave
- Shift rotation and open shifts that staff can pick up
- Timesheets and hours per job or project
- Single sign-on and two-factor sign-in for admins

EMPLOYEE APP

- Request a missed day to be added from the app
- Refinement of field tracking and the live map

Thank you · Inzivo team · October 2026